

# Product Carbon Footprint Study Report

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Product Name: Organic cotton hooded sweatshirt

Producer Name: ■■■■

Report No: [Information provided by the user]

Date: September 01, 2026

## I. General Information

**Producer Name:** ■■■■

**Address:** [Information provided by the user]

**Legal Representative:** [Information provided by the user]

**Product Name:** Organic cotton hooded sweatshirt

**Product Function:** [Information provided by the user]

**Based on the Standard:** ecoinvent - EF v3.1

## II. Quantitative Purpose

This study has two intended goals. The first one is to identify the stages of the life cycle of the hooded sweater that contribute the most to its environmental impacts. The second one is to assess the variability of those impacts depending on the production site or the behavior of the user. This LCA is carried as an internal request from GreenDelta, to provide openLCA users with a guided case LCA study.

## III. Scope of Quantification

### 1. Functional or Declaratory Units

Functional or declarative units in 1 organic cotton 2XL hooded sweater, with a weight of 750g, used for 1 year..

### 2. System Boundary

The product is assessed in a cradle-to-grave system, from the extraction of raw materials to the disposal of the used product in 2025. The extraction of raw materials and pre-processing consist of the production of the cotton fabric, the zipper, and polyester resin. This life cycle stage, as well as the manufacturing stage, take place in India. The product is then exported to Berlin where it is used and disposed.

### 3. Cut-off Criteria

[Information provided by the user]

### 4. Time Scale

[Information provided by the user]

## IV. Inventory Analysis

### 1. Description of Data Sources

**Primary Data:** [Information provided by the user]

**Secondary Data:** ecoinvent

## 2. Principles and Procedures for Allocation

**Basis of Allocation:** ■■■■

**Allocation Procedure:** [Information provided by the user]

**Specific Allocations:** [Information provided by the user]

## 3. Data Quality Evaluation (Optional)

[Information provided by the user]

## V. Impact Assessment

### 1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

### 2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of Organic cotton hooded sweatshirt produced by ■■■■, per functional unit of product, from raw material acquisition to end of life is **20.937 kg CO<sub>2</sub>-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

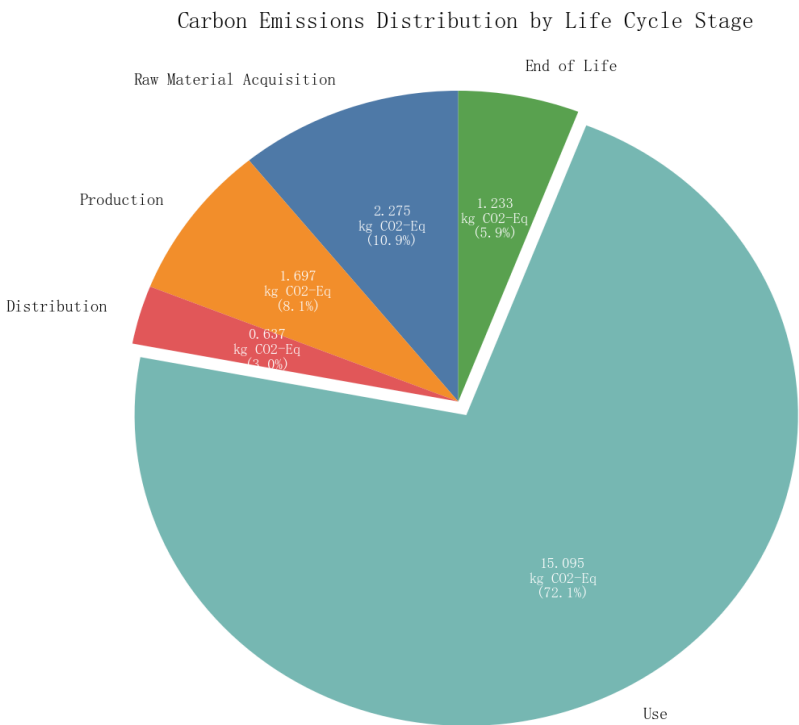


Table 1: Carbon Emissions by Life Cycle Stage

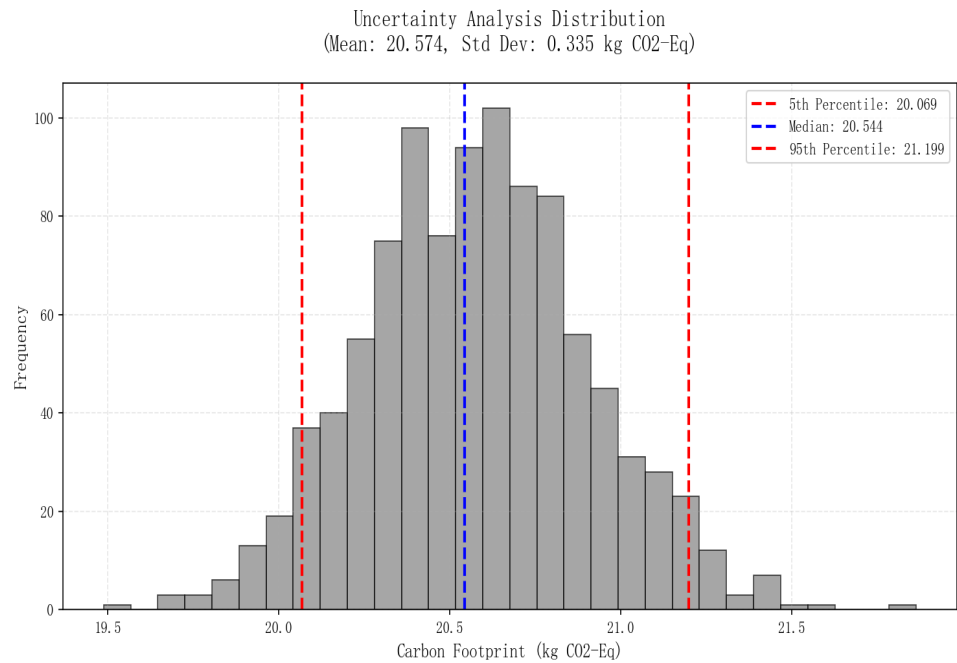
| Life Cycle Stage         | Carbon Footprint (kg CO <sub>2</sub> -Eq) | Percentage (%) |
|--------------------------|-------------------------------------------|----------------|
| Raw Material Acquisition | 2.275                                     | 10.87          |
| Production               | 1.697                                     | 8.11           |
| Distribution             | 0.637                                     | 3.04           |

|              |               |               |
|--------------|---------------|---------------|
| Use          | 15.095        | 72.10         |
| End of Life  | 1.233         | 5.89          |
| <b>Total</b> | <b>20.937</b> | <b>100.00</b> |

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



| Statistical Metric            | Value (kg CO2-Eq) |
|-------------------------------|-------------------|
| Mean                          | 20.574            |
| Standard Deviation            | 0.335             |
| Coefficient of Variation (CV) | 1.63%             |
| Median                        | 20.544            |
| 5th Percentile                | 20.069            |
| 95th Percentile               | 21.199            |

## VI. Interpretation of Results

### 1. Description of Results

The total carbon footprint of the organic cotton hooded sweatshirt is 20.937 kg CO<sub>2</sub>-Eq, representing a moderate impact level for apparel products when benchmarked against industry averages for similar garments. This value reflects a full cradle-to-grave assessment and highlights the product's environmental profile under typical consumer usage patterns.

Lifecycle contributions reveal a highly skewed distribution, with the use phase dominating at 72.1% (15.095 kg CO<sub>2</sub>-Eq). Raw material extraction follows at 10.9%, while production, end-of-life, and distribution contribute 8.1%, 5.9%, and 3.0% respectively. This pattern indicates that consumer-phase activities, primarily laundering and drying, drive the majority of emissions.

The primary emission source is clearly the use stage, underscoring the influence of energy-intensive maintenance behaviors over the product's lifespan. Secondary hotspots include raw material cultivation and manufacturing processes, suggesting opportunities for targeted interventions beyond the dominant use-phase impact.

### 2. Statement of Assumptions and Limitations

Key assumptions in this study include average consumer washing frequency of 20-30 cycles per year, standard grid electricity mixes for drying, and typical product lifespan of three to five years. Transportation distances and end-of-life disposal rates are modeled using regional averages rather than product-specific logistics data.

Data sources rely on secondary databases such as Ecoinvent and industry reports, which may introduce variability due to generic cotton cultivation parameters and aggregated energy consumption figures. Quality limitations arise from potential temporal mismatches in emission factors and lack of primary site-specific measurements for the producer's operations.

System boundaries are restricted to cradle-to-grave stages, excluding upstream capital goods, consumer transport to retail, and indirect land-use changes. The assessment omits biogenic carbon sequestration credits and potential microplastic releases during use, narrowing the scope to direct greenhouse gas emissions only.

### 3. Suggestions for Improvement

- Target the use phase by developing care labels that recommend cold-water washing and air-drying, potentially reducing emissions by 40-60% through consumer education campaigns.
- Shift raw material sourcing to low-impact organic cotton variants with verified regenerative farming practices to lower the 10.9% contribution from cultivation.
- Implement renewable energy in production facilities to cut the 8.1% manufacturing share, focusing on solar-powered spinning and dyeing operations.
- Enhance end-of-life recycling partnerships to improve the 5.9% disposal impact via take-back programs that divert textiles from landfill.

- Improve data quality by collecting primary energy and transport data from suppliers, supplemented with sensitivity analyses for use-phase assumptions.

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